

Test 2

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1 Neural Network

a neuron takes inputs $x_1=2$ and $x_2=1$ and the weights are $w_1=0.4$ and $w_2=0.6$ and bias $b=-0.2$

$$z = 2(0.4) + 1(0.6) + (-0.2)$$

$$z = 1.2$$

the sigmoid function is

$$\sigma(z) = 1/(1 + e^{-z})$$

output = 0.7685

if the weights are doubled the output of the sigmoid function would change to 0.9309

2 inferencing

if a model 2 GB and we have 8 GB of space available then we could have 4 such models fit in the memory

batching requests is efficient because in batching the model takes the incoming requests while processing other requests. this assures that there is no stalling while the model is working.